

Advanced Natural Language Processing

AI Agents – Introduction

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AI Agents

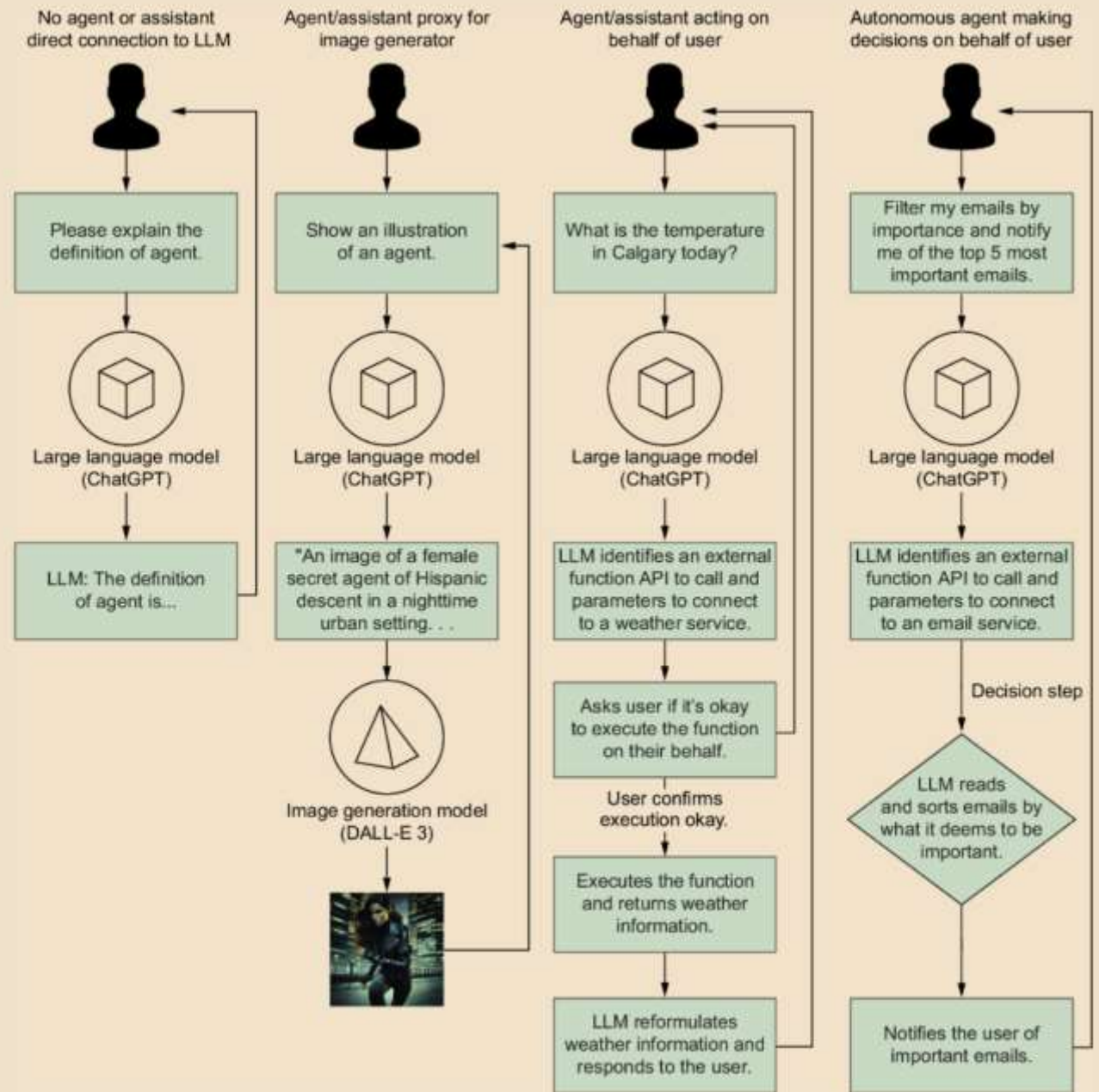


Figure 1.1 The differences between the LLM interactions from direct action compared to using proxy agents, agents, and autonomous agents

AI Agents

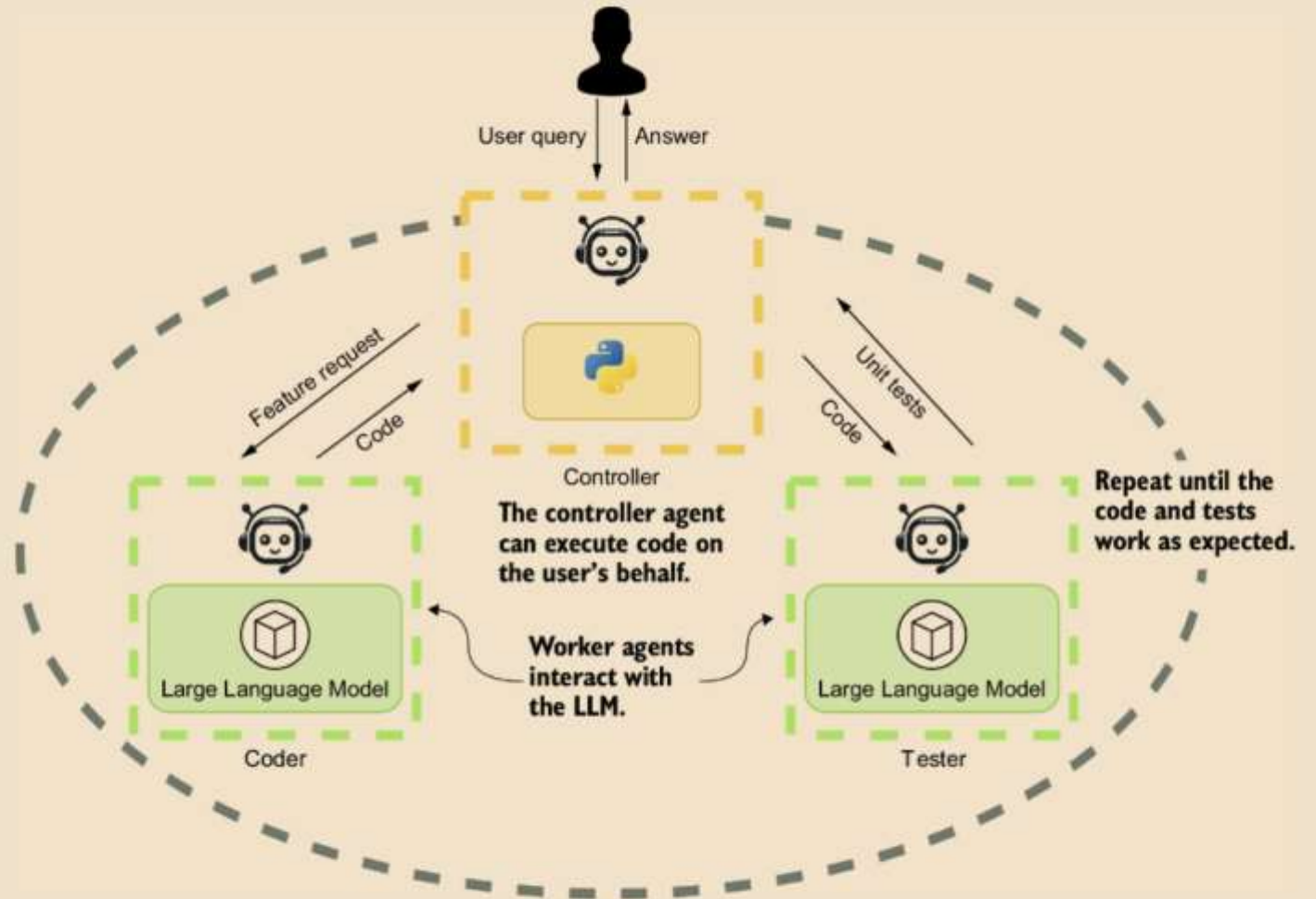


Figure 1.2 In this example of a multi-agent system, the controller or agent proxy communicates directly with the user. Two agents—a coder and a tester—work in the background to create code and write unit tests to test the code.

Component of an AI Agent

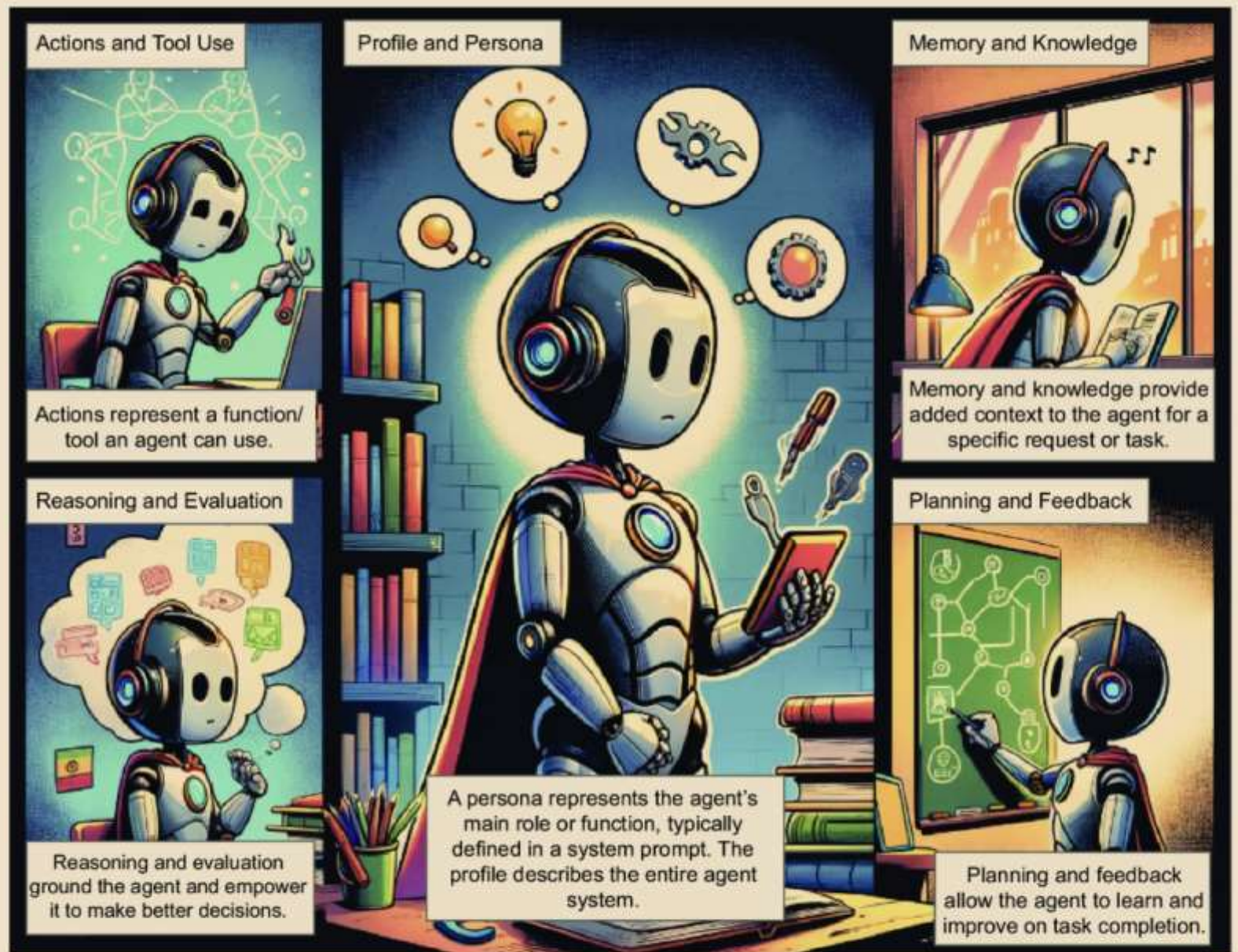


Figure 1.3 The five main components of a single-agent system (image generated through DALL-E 3)

Agent Profile



Agent persona: We'll understand how to clearly define the persona, specifying their role and characteristics to guide the agent effectively.

Agent role and demographics: We'll see how relevant demographic and role details can provide agent context, such as age, gender, or background, for a more relevant interaction.

Human vs. AI assistance for persona generation: We'll highlight the role of human involvement in persona generation, whether it's entirely human driven or assisted by LLMs or other agents.

Innovative persona techniques: Prompts generated through data or other novel approaches such as evolutionary algorithms to enhance agent capabilities.

Figure 1.4 An in-depth look at how we'll explore creating agent profiles

Action targets: We'll learn the importance of defining action targets, whether for task completion, exploration, or communication, to clarify the agent's objectives.

Action space and impact: We'll learn the significance of understanding how actions affect task completion and their effect on the agent's environment, internal states, and self-knowledge.

Action generation methods: We'll see the various ways actions can be generated, such as manual creation, memory recollection, or plan following, to illustrate the diversity of agent behaviors.

Action and Tool Use



Action Target

Semantic or native functions

Action Space

Tools, self-knowledge, other agents

Action Impact

Environments, new actions, internal states, other agents

Action Generation

Manual, memory recollection, plan following



Agent Actions
& Tools



Memory and Knowledge



Retrieval Structure

- Unified
- Hybrid

Retrieval Formats

- Language
- Databases
- Embeddings
- Lists

Retrieval Operation

- Augmentation
- Semantic Extraction
- Compression

Retrieval structure variety: We'll learn about the diverse memory structures agents can employ, including unified and hybrid approaches, enabling flexibility in information storage.

Retrieval formats: We'll explore the various data sources for memory, such as language (e.g., PDF documents), databases (relational, object, or document), and embeddings, offering a rich pool of information to draw upon.

Semantic similarity: We'll learn how embeddings enable semantic similarity searches, facilitating efficient retrieval of relevant data and enhancing the agent's decision-making capabilities.

Agent Memory & Knowledge



Reasoning

- Zero-shot prompting
- One-shot prompting
- Few-shot prompting
- Chain of thought
- Tree of thought
- Skeleton of thought

Evaluation

- Self-consistency
- Prompt chaining

Reasoning enables the agent to self-reflect and internally reason out the completion of a task or tasks.

Evaluation provides the basis for an agent's self-reflection on working through and upon task completion.

Agent Reasoning &
Evaluation

We'll look at various planning strategies with and without feedback—from basic and sequential planners to automatic tool use with reasoning.

Feedback may come from a variety of sources, such as environmental, human, and an LLM via various constructive feedback patterns.

Planning and Feedback

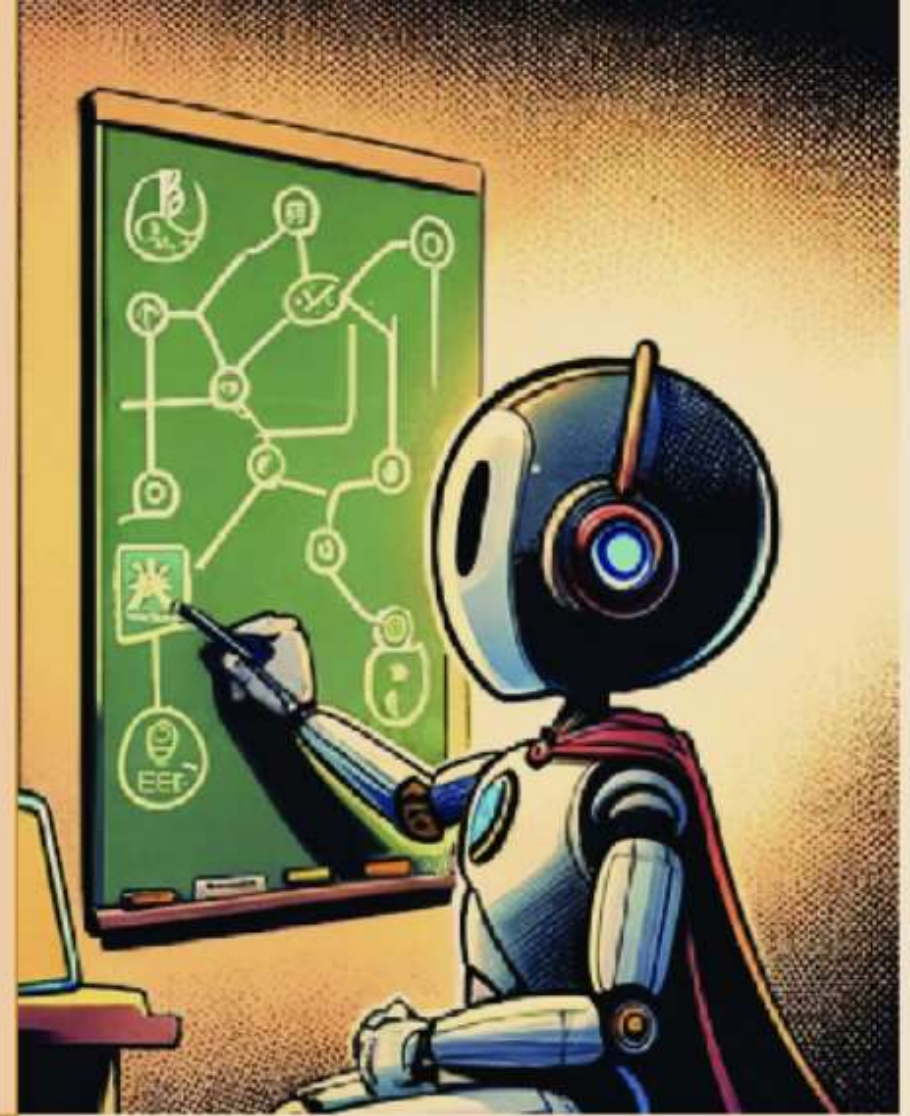


▶ Planning without feedback (autonomous)

- Basic planning
- Automatic reasoning with tool use
- Sequential planning

▶ Planning with feedback

- Environmental feedback
- Human feedback
- LLM feedback
- Adaptive constructive feedback



Agent Planning &
Feedback

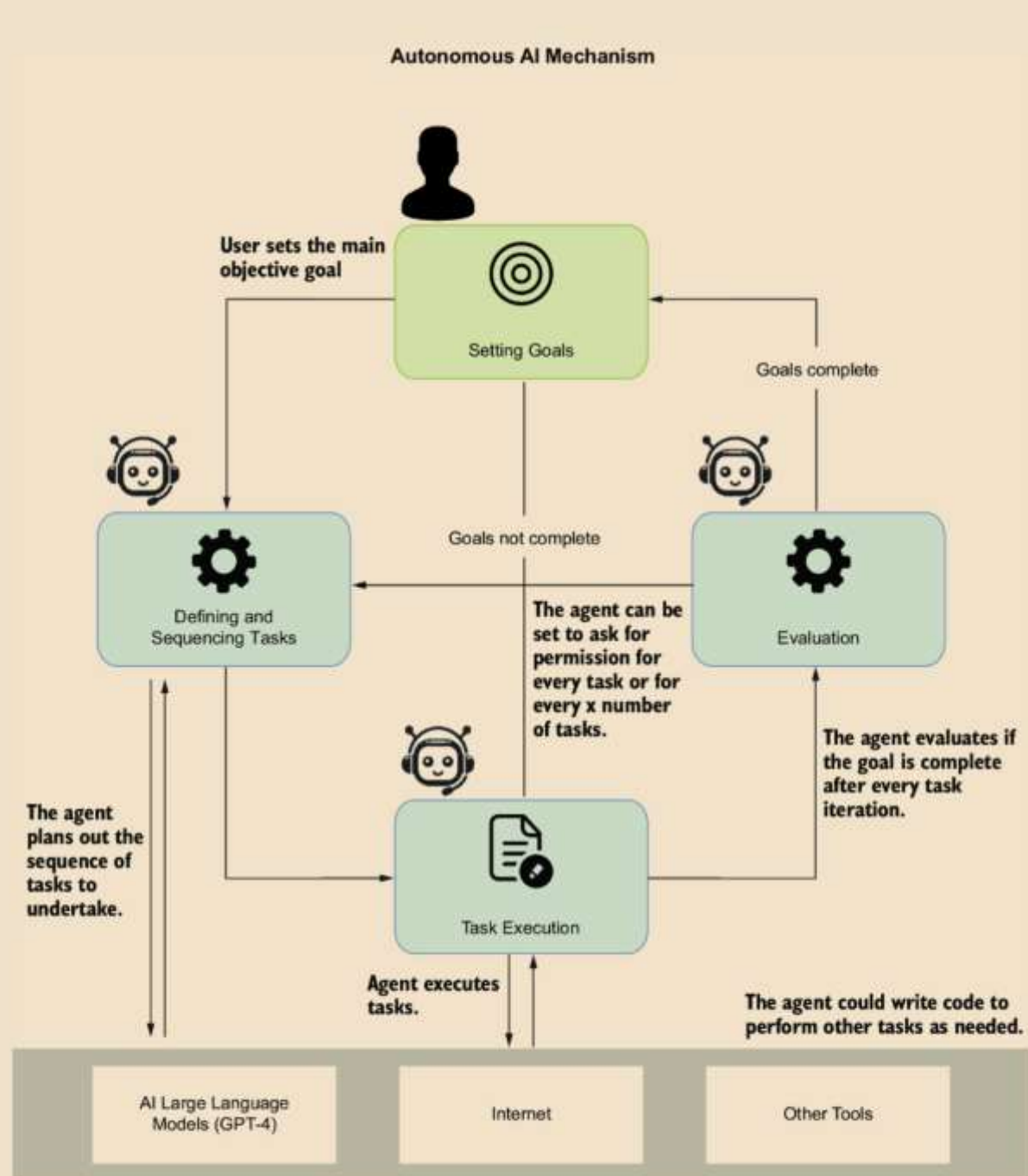
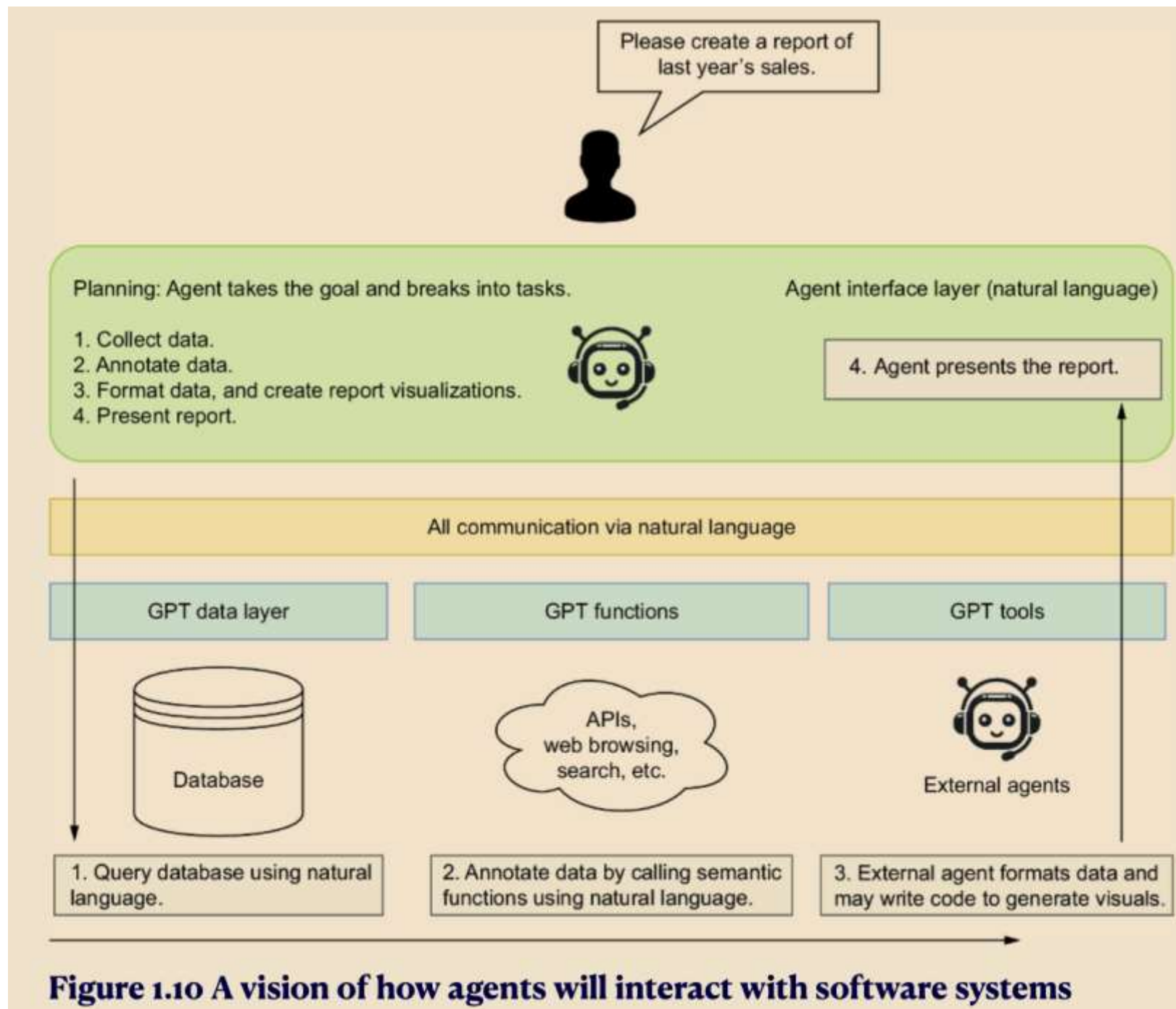


Figure 1.9 The original design of the AutoGPT agent system



- An agent is an entity that acts or exerts power, produces an effect, or serves as a means for achieving a result. An agent automates interaction with a large language model (LLM) in AI.
- An assistant is synonymous with an agent. Both terms encompass tools such as OpenAI's GPT Assistants.
- Autonomous agents can make independent decisions, and their distinction from non-autonomous agents is crucial.

- The four main types of LLM interactions include direct user interaction, agent/ assistant proxy, agent/assistant, and autonomous agent.
- Multi-agent systems involve agent profiles working together, often controlled by a proxy, to accomplish complex tasks.
- The main components of an agent include the profile/persona, actions, knowledge/memory, reasoning/evaluation, and planning/feedback.
- Agent profiles and personas guide an agent's tasks, responses, and other nuances, often including background and demographics.

- Actions and tools for agents can be manually generated, recalled from memory, or follow predefined plans.
- Agents use knowledge and memory structures to optimize context and minimize token usage via various formats, from documents to embeddings.
- Reasoning and evaluation systems enable agents to think through problems and assess solutions using prompting patterns such as zero-shot, one-shot, and few-shot.
- Planning/feedback components organize tasks to achieve goals using single-path or multipath reasoning and integrating environmental and human feedback.
- The rise of AI agents has introduced a new software development paradigm, shifting from traditional to natural language-based AI interfaces.

- Understanding the progression and interaction of these tools helps develop agent systems, whether single, multiple, or autonomous.